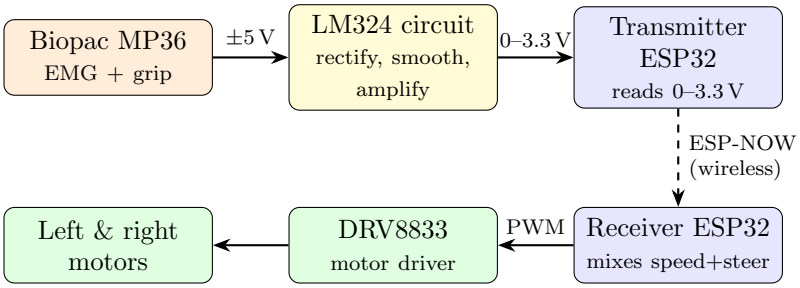
EMG-Controlled Tank Robot

Student Build Guide

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The idea: your muscle signal (EMG) controls the robot's *speed* and your grip force controls its *steering*.
The how: A Biopac MP36 measures both, a small circuit cleans up the signals, one ESP32 sends them over WiFi, and a second ESP32 on the robot turns them into motor commands.
Build order: Read the safety rules → Transmitter circuit → Tank Robot → Biopac setup → Pre-power checks → Upload code → Test one piece at a time.

1 The Big Picture



ESP-NOW is a direct board-to-board WiFi link: no router, no internet, very low delay. You only need to tell the transmitter the receiver's MAC address (Section 11).

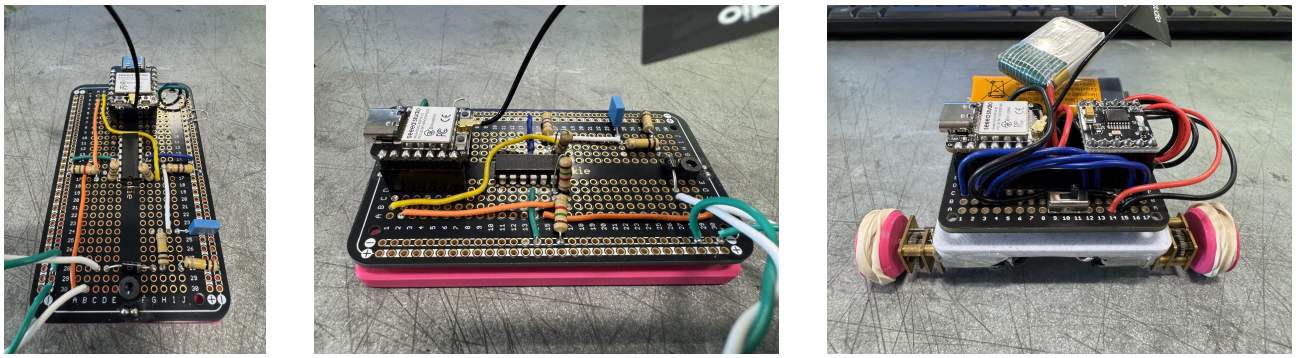


Figure 1: Left and middle: finished transmitter (XIAO ESP32-S3 + LM324); the twisted green/white pairs go to the two Biopac outputs. Right: finished robot (XIAO ESP32-S3 + DRV8833, LiPo and 9 V battery).

2 Before You Build: Four Rules

▲ Read this first

1. **Never connect the Biopac directly to the ESP32.** The Biopac output can swing to ±5 V; the ESP32 pins survive only 0–3.3 V. The LM324 circuit is what keeps the microcontroller safe.
2. **All grounds must connect together.** Biopac ground, ESP32 GND, LM324 pin 11, LiPo −, and motor battery −. Missing grounds cause random behavior.
3. **Parts have a direction.** The diode stripe is the cathode and must face the op-amp. The LM324 notch/dot marks pin 1. Check both *before* soldering.
4. **Never solder a module directly to the board.** The ESP32's and DRV8833 plug into header sockets in case you need to troubleshoot (Section 6).

New to building circuits?

- **Resistor color codes:** brown–black–yellow = 100 kΩ. When unsure, measure with a multimeter.
- **Breadboard/perfboard rows:** holes in the same numbered row (A–E, or F–J) are connected; the long +/− strips on the edges are the power rails.
- **Build one channel, test it, then build the next.** Finding a mistake in 5 parts is easy; not so much in 15.
- **Use the continuity (beep) mode** on your multimeter to confirm every ground and every connection after soldering. New to multimeters? Read SparkFun's How to Use a Multimeter; the voltage, resistance and continuity sections will be the most relevant.
- **Never soldered before?** Read the Adafruit Guide to Excellent Soldering first. It shows what good and bad joints look like and how to fix a mistake with solder wick. Then practice 10–20 joints on a scrap piece of perfboard before touching your real board.
- **Common beginner mistakes:** LM324 inserted backward, a resistor one row off, and solder bridges between neighboring pads. Look for all three before powering up.

3 Power and Safety

3.1 What powers what?

Board	Power source	Connects to
Transmitter ESP32	Laptop USB-C	USB-C port
LM324	Transmitter ESP32	ESP32 **5V (VBUS)** → LM324 pin 4
Receiver ESP32	3.7 V LiPo	**BAT+ / BAT− pads on the back** of the ESP32 (Figure 2)
Motors	9 V battery (or 4×AA)	DRV8833 **VCC** and GND only

Why the BAT pads? They feed the XIAO's built-in charger, so the LiPo charges whenever the robot's ESP32 is plugged into USB-C (the CHARGE LED lights while charging). Connecting a LiPo

to the 5V/VBUS pin instead bypasses the charger, and if USB is plugged in at the same time, USB power is pushed straight into the battery, ignoring the battery management.

3.2 LiPo battery rules

LiPo cells store a lot of energy in a soft pouch. Treated well they are safe. Shorted, punctured, or overcharged they can **swell, smoke, or catch fire**.

▲ LiPo safety

1. **Check polarity with a multimeter before connecting.** Red is not always +; connector wiring varies between vendors. Reversing a LiPo will destroy the ESP32.
2. **Only connect to the BAT+/BAT− pads.** Never to 5V/VBUS, 3V3, or the motor driver.
3. **Charge only through the ESP32's USB-C port**, and never leave a charging battery unattended.
4. **Never short the leads.** Cover exposed wire ends with tape or heat-shrink; do not let the two leads touch, and do not cut both wires at once with the same cutter.
5. **Do not puncture, crush, bend, or squeeze the pouch.** Mount it with tape or Velcro, not with screws or hot glue directly on the cell.
6. **Stop immediately if the battery is puffy, hot, or smells sweet.** Disconnect it (if safe to touch), place it in the LiPo safety bag or an open metal container, and tell the instructor. Do not charge or reuse it.
7. **Disconnect or switch off the battery** whenever you are wiring, soldering, or done for the day.

3.3 Other safety notes

- **Power off before you rewire.** Turn the robot's switch off (or unplug the batteries) before moving any wire. A loose wire touching the wrong pin while powered is the fastest way to kill a board.
- **Motor battery:** disconnect it when you are not driving. A 9 V battery drains quickly under motor load; a dying battery can look like a code bug.
- **Soldering:** wear safety glasses, work with ventilation or a fume extractor, keep the tip clean during , and wash your hands afterward.
- **Subject safety:** electrodes connect *only* to the Biopac. The MP36 electrically isolates the person from the analog output. Never connect electrode leads to your circuit.

4 Parts List

Qty	Part	Notes
2	Seeed XIAO ESP32 (S3, C3 or C6)	Use an **S3 on the robot**. Other boards can serve as alternative with some modifications.

1	Biopac MP36 + BSL PRO	Two MP36 boxes: one for EMG, one for the grip sensor.
1	EMG leads (SS2LB) + electrodes	Speed input.
1	Hand dynamometer (SS25LA or SS25LB)	Steering input (almost any Biopac sensor works too).
1	DRV8833 motor driver module	Drives both motors.
2	Geared DC motors + wheels	Mounted on the 3D-printed chassis.
1	3D-printed chassis parts	STL files are in the repository's `stl/` folder.
1	Slide switch	Robot power on/off.
1	LM324 quad op-amp	One chip handles both channels.
1	1N4007 diode	EMG channel rectifier.
2	100 kΩ resistors	EMG channel: bleeder + low-pass filter.
1	100 nF capacitor	EMG channel low-pass filter.
2 + 2	R_a and R_f resistors	Sets the gain; one pair per channel (Section 7.3).
2 sets	Female header sockets (2.54 mm)	Board side: one pair per XIAO and per DRV8833.
1	3.7 V LiPo	Powers the receiver ESP32.
1	9 V or 4×AA batteries	Powers the motors only.

Also: perfboard or breadboard, hookup wire, **data-capable** USB-C cables, multimeter. A bench power supply and oscilloscope are helpful for testing.

5 Know Your Pins

Before wiring anything, understand how components will be connected. Every connection in this guide refers to one of these diagrams.

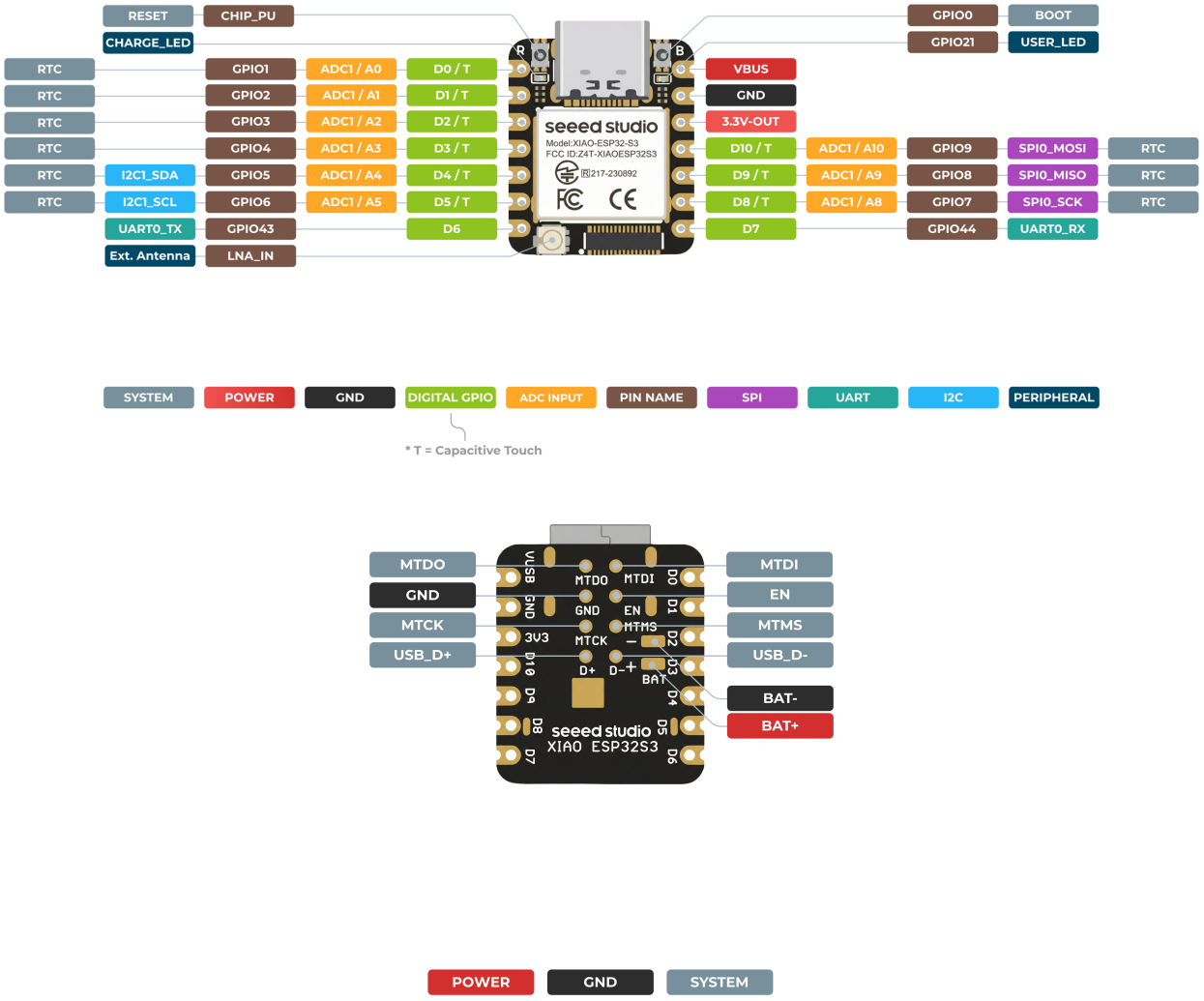
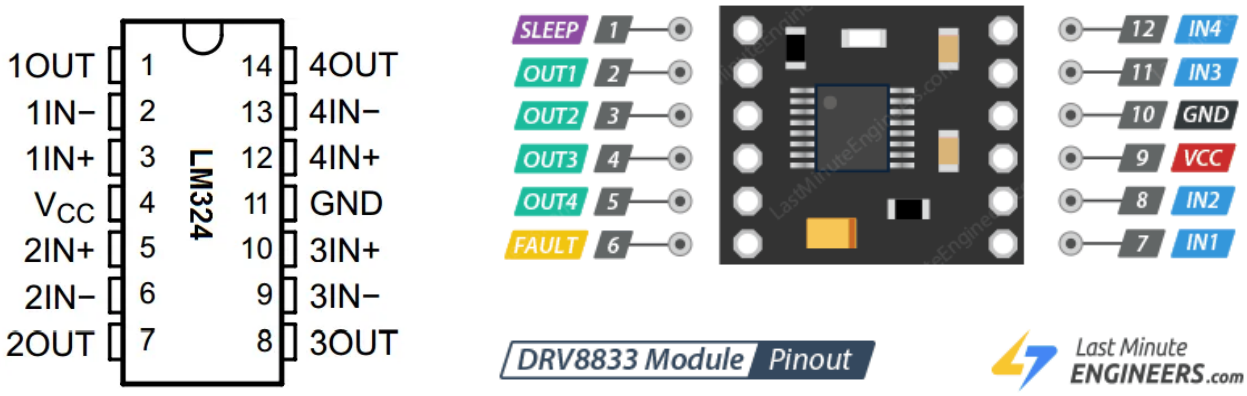


Figure 2: XIAO ESP32-S3 pinout, front (top) and back (bottom). A0–A3 are the top four pins on the left (D0–D3). VBUS (5 V), GND and 3V3 are the top three on the right. The **BAT+/BAT−** pads are on the back. B = boot button, R = reset button. (Images: Seeed Studio.) For the C3 or C6, see the Seed wiki links at the end of this guide.



(a) LM324, top view (b) DRV8833 module (image: LastMinuteEngineers)

Figure 3: (a) Find the notch or dot: pin 1 is to its left, and pins count down the left side and back up the right. This guide uses op-amps 1 (pins 1–3) and 4 (pins 12–14). (b) The module labels motor power as **VCC** (this is the chip's VM). Leave **SLEEP** and **FAULT** unconnected; SLEEP is pulled high on this module so the driver stays awake.

6 Use Headers, Not Solder

The XIAO boards and the DRV8833 are **shared lab equipment**. They go back in the kit at the end of the semester and the next group builds with them. So they **plug into** your board instead of being soldered to it.

On the module (male pins)

- XIAO and DRV8833 come with (or get) male header pins soldered on.
- If yours already has pins, **do not remove or resolder them.**

On your board (sockets)

- Solder **female header sockets** where each module sits.

Why it matters to you, too:

- A dead or suspect board can be swapped in seconds instead of desoldered.
- You can pull the ESP32 out to test your circuit safely (Section 10, step 3).
- A chip inserted backward in a socket can be flipped; one soldered in backward usually cannot be saved.

Getting headers straight

1. **Use a module as a jig:** plug the female sockets onto the module's pins, place the whole stack on your board, and solder the sockets from underneath. They will line up exactly.
2. **Tack one corner pin first,** check that the socket sits flat and square, reheat that pin to adjust if needed, then tack the opposite corner. Solder the rest only once it is straight.
3. **Remove the module before soldering the remaining pins** if you can, so its pins and chip are not heated.
4. **Solder wires to the board, not to the module.** Run every connection to a socket pin's pad on the board side.

The one exception: the LiPo connects to the XIAO's BAT pads. Solder jumper wires to the back of the ESP32 to connect to the header pins. A switch can be places inline to interrupt the connection to the battery.

End of lab: unplug every module, and return them to the lab.

7 Build the Transmitter Circuit

7.1 What the circuit does

The two channels are handled differently because the signals are different:

	Channel 1: EMG → Speed	**Channel 2: Grip → Steering**
Raw signal	Fast, oscillates above *and* below 0 V	Slow, already positive
Processing	Rectify → smooth → amplify	Amplify only
LM324 pins	1 (out), 2 (−), 3 (+)	14 (out), 13 (−), 12 (+)
ESP32 pin	A0	A1

Why each EMG step is needed

1. **Rectify (diode):** EMG averages to zero. The diode passes only the positive half of the signal.
2. **Smooth (RC low-pass, 100 kΩ + 100 nF):** turns the spiky signal into an *envelope* that follows muscle effort. Cutoff $f_c = \dfrac{1}{2\pi RC} = \dfrac{1}{2\pi(100\,\text{k}\Omega)(100\,\text{nF})} \approx 16\,\text{Hz}$.
3. **Bleed (100 kΩ to GND):** gives the capacitor a way to discharge. Without it, the voltage stays "stuck high" after you relax.
4. **Amplify (op-amp):** scales the envelope so a strong contraction reaches about 3 V, using the full ADC range.

Note: the 1N4007 drops about 0.6 V, so EMG smaller than that never gets through. This acts as a built-in "dead zone" at rest. If strong contractions barely register, increase the signal level in BSL PRO.

7.2 Schematic

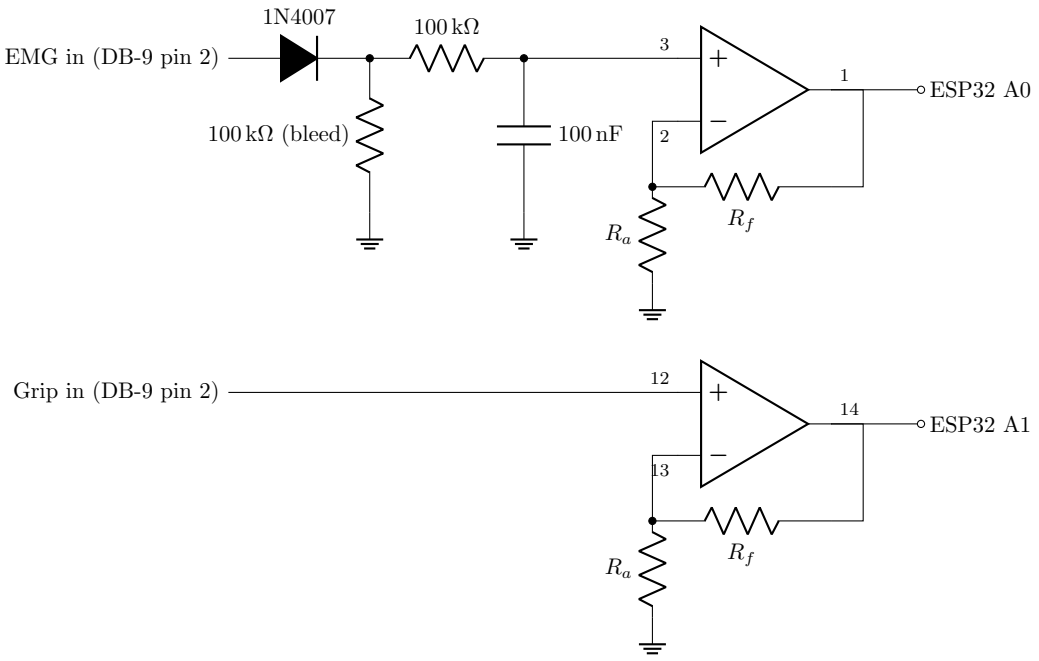


Figure 4: Transmitter conditioning circuit (one LM324). Top: EMG/speed channel. Bottom: grip/steering channel.

The full circuit is shown in Figure 4. **Power the LM324:** pin 4 → ESP32 **5V (VBUS)**, pin 11 → GND. Both Biopac DB-9 **pin 3** (ground) wires go to the GND rail. Pins 5–10 are unused spare op-amps.

Why 5 V and not 3.3 V for the LM324?

The LM324's output cannot reach its supply voltage. On 5 V it tops out near 3.5 V (safe for the ESP32 and uses most of the range). On 3.3 V it tops out near 1.8 V, so you would lose about half your resolution.

7.3 Choosing the gain

This is a non-inverting amplifier:

$$G = 1 + \frac{R_f}{R_a} \qquad \text{Target: } G \approx \frac{3.0\ \text{V}}{V_{\text{peak}}}$$

where V_{peak} is the largest voltage at the op-amp + input during a strong contraction. A multimeter averages too slowly to catch EMG peaks, so measure it with an **oscilloscope** (new to scopes? see SparkFun's How to Use an Oscilloscope or watch the Oscilloscope Basics for Beginners).

Example: $V_{\text{peak}} = 0.5\,\text{V} \Rightarrow G \approx 6 \Rightarrow R_f/R_a \approx 5 \Rightarrow R_a = 10\,\text{k}\Omega$, $R_f = 47\,\text{k}\Omega$ ($G = 5.7$). Too little gain: the robot barely moves. Too much: it saturates and acts like an on/off switch.

8 Build the Robot

Chassis: print the parts from the repository's `stl/` folder. Mount the two geared motors and the front ball caster, wrap the wheels with rubber bands for grip, and attach the perfboard, LiPo and motor battery on top (Figure 5). **You are also encouraged to design your own chassis.**

Wiring (DRV8833 module pin numbers from Figure 3b):

ESP32 → DRV8833

ESP32	DRV8833	Job
A0	IN1 (pin 7)	Left speed (PWM)
A1	IN2 (pin 8)	Left direction
A2	IN3 (pin 11)	Right speed (PWM)
A3	IN4 (pin 12)	Right direction
GND	GND (pin 10)	Common ground

Power & motors

From	To
9 V or 4 AA +	DRV8833 VCC (pin 9)
9 V or 4 AA −	DRV8833 GND (pin 10)
LiPo +	XIAO **BAT+** pad
LiPo −	XIAO **BAT−** pad
Left motor	OUT1, OUT2 (pins 2, 3)
Right motor	OUT3, OUT4 (pins 4, 5)

Two batteries on purpose: motors create electrical noise and voltage dips that can reset the ESP32. A separate motor battery keeps the brain and the muscles apart. **The grounds still must connect.**

Motor spins backward? Swap its two wires at the DRV8833 output or swap the wires at the motor end.



Figure 5: Left: DRV8833 (front) and ESP32-S3 (back) stacked on the robot's perfboard. Right: underside with both geared motors and the front ball caster.

9 Set Up the Biopac

Electrodes (EMG):

1. Clean the skin with an alcohol wipe; let it dry.
2. Place two electrodes on the muscle belly, about 5 cm apart, along the fiber direction.
3. Place the reference (ground) electrode on a bony spot (wrist, elbow, or knee).

BSL PRO:

1. Box 1, Channel 1 → EMG (5–1000 Hz). Box 2, Channel 1 → Hand Dynamometer preset.

2. **MP36 menu → Output Control**: assign Channel 1 to the analog output and enable it (on both boxes).
3. Connect each box's rear DB-9: **pin 2 = signal**, **pin 3 = ground**.

▲ Most common "it doesn't work"

The analog output is **only live while BSL PRO is recording.** Press record before testing.

Changing Biopac Sensor

EMG for speed is a requirement of this project. The second sensor can be your choice. Pay attention to which sensor you use because that could require a change in circuit and/or BSL Pro settings.

10 Before You Plug Anything In

Do these checks **in order**. Each one takes a minute and protects parts that much longer to replace.

1. Look (power off).

❑ LM324 notch matches the diagram; pin 1 is where you think it is
❑ Diode stripe faces the op-amp
❑ No solder bridges between neighboring pads; every joint is shiny and cone-shaped
❑ Every module is fully seated and facing the right way

2. Beep (multimeter continuity, power off).

❑ 5V rail to GND: **must NOT beep.** If it beeps you have a short; find it before going further
❑ Every ground beeps to every other ground (ESP32 GND, LM324 pin 11, both DB-9 pin 3, DRV8833 GND, LiPo −, motor battery −)
❑ LM324 pin 1 beeps to ESP32 A0; pin 14 beeps to ESP32 A1

3. Measure the transmitter output with the ESP32 removed.

1. Take the ESP32 out of its socket. Power the 5V rail from a bench supply set to **5.0 V** (current limit about 50 mA). You can also power it from the ESP32 using a female to female jumper wire.
2. Start a BSL PRO recording and contract hard / squeeze hard.
3. Measure LM324 pins 1 and 14 to GND. **Both must stay below 3.3 V.** If not, lower the gain before going further.
4. Power off, then put the ESP32 back in.

4. Robot: power up in stages, wheels off the ground.

1. Check the LiPo polarity with the multimeter, then connect it to the BAT pads (Section 3).
2. First power the receiver from USB only, with the motor battery disconnected. Confirm it prints its MAC address in the Serial Monitor.
3. Connect the motor battery last, with the power switch off. Switch on with the wheels lifted off the table.

11 Upload the Code

One-time setup (Arduino IDE):

1. **File** → **Preferences**, add this Boards Manager URL:
   https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
2. **Tools** → **Board** → **Boards Manager**, install *esp32 by Espressif Systems*.
3. Select your board (`XIAO_ESP32S3`, `XIAO_ESP32C3`, or `XIAO_ESP32C6`) and its COM port.

Pair the two boards (order matters):

1. Upload `EMGCarReceiver.ino` to the **robot** ESP32.
2. Open Serial Monitor at **115200 baud**, press reset, and copy the printed MAC address, e.g. `AA:BB:CC:DD:EE:FF`.
3. In `EMGCarTransmitter.ino`, replace the placeholder with your address:

```
uint8_t receiverMAC[] = {0xAA, 0xBB, 0xCC, 0xDD, 0xEE, 0xFF};
```

4. Upload the transmitter code. The boards are now paired.

Can't upload?

1. **No COM port appears:** your USB-C cable may be charge-only. Try a different cable (data-capable), and a different USB port.
2. **Port appears but upload fails or times out:** put the board in bootloader mode. Unplug it, **hold the B (boot) button**, plug it back in, then release B. Upload again. When it finishes, press **R** (reset) to run your code.
3. **Wrong board selected:** check **Tools** → **Board** matches the chip printed on your XIAO (S3, C3 or C6).
4. **"Port busy" / access denied:** close any other Serial Monitor or program using that port.
5. **Serial Monitor is blank on the S3:** set **Tools** → **USB CDC On Boot** → **Enabled**, re-upload, then press R.

11.1 How the code works

- **Transmitter:** reads A0 (speed) and A1 (steering) as numbers from 0 (0 V) to 4095 (3.3 V), packs them, and sends them about every 250 ms.
- **Receiver:** unpacks the numbers and calculates left and right PWM values. The steering value is centered around zero; the robot turns by giving one side more power than the other:

$$\text{Left} = \text{Speed} - \text{Steer}, \qquad \text{Right} = \text{Speed} + \text{Steer}$$

  Each result is scaled to 0–255 (PWM) and sent to the DRV8833. If no packet arrives for 5 s, the motors stop.

Both files share this packet. **If you change it, change it in both files and re-upload both.**

```
struct ControlPacket {
  uint16_t SensorVal1;   // Speed:    0 -> 4095
  uint16_t SensorVal2;   // Steering: 0 -> 4095
};
```

Where to edit: blocks marked `// %%%%%` are the communication framework (be careful with what you change since it can cause connection issues). Blocks marked `// =====` are areas where you can add your code.

11.2 Tuning knobs

Variable	File	Default	Effect
`EMGcap`	Receiver	–	Speed threshold. Raise if motors run at rest; lower if they never start.
`maxSteeringValue`	Receiver	100	Grip needed for a full turn. Raise if it turns too easily. Note: turning can also be affected by your design.
`alpha`	Transmitter	0.5	Smoothing (0–1). Higher = faster but jittery; lower = smooth but laggy.
`delay(250)`	Transmitter	250 ms	Update rate. 50 ms feels snappier but the ESP32 runs hot.

12 Test One Piece at a Time

Use the **Serial Monitor** (115200 baud) to read values, and the **Serial Plotter** (Tools menu) to *see* them as live graphs. The plotter is the fastest way to tell whether your signal looks right.

1. **Transmitter alone:** relaxed muscle should read roughly 0–400; contracting should push values up toward ~3700. You should see `Sent`.
2. **Receiver on the bench, wheels off the ground:** you should see `Left: # , Right: #` change as you contract and squeeze.
3. **On the floor.** Tune `EMGcap` and `maxSteeringValue` until it drives the way you want.

12.1 Troubleshooting

Symptom	Check / Fix
Reading always 0	BSL PRO recording? Analog output assigned and enabled? DB-9 pins 2/3 correct?
Reading stuck high at rest	Bleeder resistor missing, or LM324 on 3.3 V instead of 5 V.
Max reading too low	LM324 on 3.3 V, or gain too small (increase R_f).
`FAILED` on transmitter	Wrong MAC address, or receiver not powered.

Motors never move	`EMGcap` too high; check receiver Serial Monitor shows values; motor battery connected and switch on?
Motors run while relaxed	`EMGcap` too low.
Steering does nothing	Channel 2 wiring to A1; dynamometer set up and output enabled on Box 2.
Robot resets when motors start	Motor and ESP32 sharing a battery, or loose ground.
Robot slows down over time	Motor battery is dying; replace it.
Board gets hot right away	Short circuit or reversed power. **Unplug immediately** and recheck Section 10.

13 Make It Yours

The circuit and code only care about a *voltage* arriving at A0/A1, so almost anything is fair game once the base system works:

- **New sensors:** replace the dynamometer with a second EMG channel, or any MP36-compatible sensor. Check the signal on an oscilloscope first.
- **More channels:** the LM324 has two spare op-amps (pins 5–10). Each extra channel needs its own Biopac box and a new field in `ControlPacket`.
- **True tank drive:** one EMG per side, each controlling its own motor.
- **Reverse:** use a third signal; set the direction pin `HIGH` and the speed pin to 0.
- **Adjustable gain:** replace R_f with a potentiometer.
- **Feedback LED:** drive the onboard LED brightness from the speed value (not on the C3).
- **New chassis:** edit the STL files or design your own.

Quick Reference

Transmitter	
A0	LM324 pin 1 (speed)
A1	LM324 pin 14 (steering)
5V (VBUS)	LM324 pin 4
GND	LM324 pin 11, DB-9 pin 3 (both)

Receiver	
A0 / A1	DRV8833 IN1 / IN2 (left)
A2 / A3	DRV8833 IN3 / IN4 (right)
BAT+ / BAT−	LiPo + / − (back pads)
GND	DRV8833 GND, all battery −

Board notes: S3 is the primary board. C3 and C6 have no A3 (so use them as the transmitter) and the C3 has no user LED. A0/A1 work with WiFi on all three.

Links: XIAO S3 • XIAO C3 • XIAO C6 • DRV8833 tutorial • MP36 manual • Adafruit soldering guide • SparkFun oscilloscope tutorial• Oscilloscope Basics for Beginners